

6.3–6.5 Matrix Exponentials, Symmetric Matrices, and Positive Definiteness

Concise exam revision notes for continuous systems, symmetric eigenvectors, and positive definite tests.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Eigenvectors decouple matrix problems. For symmetric matrices, the decoupling is especially clean because the eigenvectors can be chosen orthonormal.

The same eigenvalue idea controls:

$$\frac{du}{dt} = Au, \quad S = Q\Lambda Q^T, \quad x^T Sx.$$

2 Differential Equations

For the system

$$\frac{du}{dt} = Au,$$

if A has independent eigenvectors $x_1, \dots, x_n$, then the solution has the form

$$u(t) = c_1 e^{\lambda_1 t} x_1 + \dots + c_n e^{\lambda_n t} x_n.$$

The constants come from the initial condition:

$$Sc = u(0), \quad c = S^{-1}u(0).$$

In matrix form,

$$u(t) = e^{At}u(0), \quad e^{At} = Se^{\Lambda t}S^{-1}.$$

Here

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{bmatrix}.$$

3 Continuous Stability

Eigenvalues of A	Behavior of e^{At}
All $\operatorname{Re} \lambda < 0$	Decays to zero.
Some $\operatorname{Re} \lambda > 0$	Blows up in that direction.
One $\lambda = 0$, rest $\operatorname{Re} \lambda < 0$	Possible steady state.
Complex $\lambda = a + ib$	a controls growth/decay; b controls oscillation.

Exam memory. For powers use $|\lambda|$. For differential equations use $\operatorname{Re} \lambda$.

4 Defective Matrices

A repeated eigenvalue may not give enough eigenvectors. Then S is not invertible, so

$$A = SAS^{-1}$$

is impossible.

For a defective repeated eigenvalue, a generalized eigenvector can replace a missing eigenvector:

$$(A - \lambda I)w = x.$$

This creates terms involving $te^{\lambda t}$ in differential equations.

Scope. For exam revision, first try ordinary eigenvectors. Use generalized eigenvectors only when the matrix is defective.

5 Symmetric Matrices

For a real symmetric matrix

$$S = S^T.$$

The key facts are:

- all eigenvalues are real;
- eigenvectors for different eigenvalues are orthogonal;
- the eigenvectors can be chosen orthonormal.

Therefore a symmetric matrix factors as

$$S = Q\Lambda Q^T.$$

Equivalently,

$$S = \lambda_1 q_1 q_1^T + \cdots + \lambda_n q_n q_n^T.$$

Exam trigger. If the matrix is symmetric, look for orthogonal eigenvectors and use $Q^T = Q^{-1}$.

6 Positive Definite Matrices

For a symmetric matrix S , positive definite means

$$x^T S x > 0 \quad \text{for every } x \neq 0.$$

Equivalent tests:

Test	Positive definite condition
Eigenvalues	All $\lambda_i > 0$.
Pivots	All pivots are positive.
Leading determinants	All leading principal determinants are positive.
Quadratic form	$x^T S x > 0$ for all $x \neq 0$.
Factorization	$S = R^T R$ with independent columns in R .

For 2×2

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix},$$

the quick test is

$$a > 0, \quad ac - b^2 > 0.$$

7 Positive Semidefinite

Positive semidefinite means

$$x^T S x \geq 0 \quad \text{for every } x.$$

Then eigenvalues are nonnegative:

$$\lambda_i \geq 0.$$

It may be singular:

$$\det S = 0$$

is allowed.

Common mistake. Positive semidefinite is not the same as positive definite. Zero eigenvalues are allowed only in the semidefinite case.

8 Similarity and Transpose

Similar matrices have the form

$$B = M^{-1} A M.$$

They have the same eigenvalues because they represent the same linear map in different coordinates.

A matrix and its transpose have the same eigenvalues:

$$\det(A^T - \lambda I) = \det(A - \lambda I).$$

But their eigenvectors do not have to be the same.

9 Quadratic Forms and Axes

A symmetric matrix rotates quadratic forms into principal axes:

$$x^T S x = (Q^T x)^T \Lambda (Q^T x).$$

For an ellipse, the eigenvectors give the axis directions and the eigenvalues control the axis lengths.

10 Common Mistakes

Mistake	Correct exam habit
Using $ \lambda $ for differential equation stability.	Use $\operatorname{Re} \lambda$.
Assuming repeated eigenvalues give enough eigenvectors.	Check the eigenspace dimension.
Using $Q \Lambda Q^T$ for a non-symmetric matrix.	Reserve that clean form for symmetric matrices.
Testing positive definiteness without symmetry.	The standard tests here assume a symmetric matrix.
Confusing semidefinite with definite.	Positive definite needs strict positivity and no zero eigenvalues.
Assuming A and A^T have the same eigenvectors.	They have the same eigenvalues, not necessarily the same eigenvectors.

11 Final Exam Checklist

Checklist.

1. For $u' = Au$, find eigenvalues and eigenvectors of A .
2. Write $u(t) = \sum c_i e^{\lambda_i t} x_i$ when A has enough eigenvectors.
3. Use $Sc = u(0)$ to find constants.
4. Use $\operatorname{Re} \lambda$ to decide continuous stability.
5. Check symmetry before using $S = Q \Lambda Q^T$.
6. For a symmetric matrix, choose eigenvectors orthonormal.
7. Test positive definiteness by eigenvalues, pivots, leading determinants, or $x^T S x$.
8. Treat zero eigenvalues as semidefinite, not definite.
9. Use similarity to preserve eigenvalues, not necessarily eigenvectors.